Tech Team Recruitment Task Submission

Backend / Full-Stack Track: Campus Event Management REST API

Candidate Name: Priyam Khetan

Enrollment Number: 11216401526

Applied Track: 03 Backend / Full-Stack Development

Submission File: PriyamKhetan_TechTeamSubmission.pdf

1. PROJECT TITLE & OBJECTIVE

Campus Event Discovery & Management API: A high-performance RESTful API built to handle event publishing, student registrations, category filtering, and authentication for campus tech fests. Built with Node.js, Express, MongoDB, and JWT authentication.

2. TECHNICAL ARCHITECTURE & TECH STACK

Component	Technology Choice	Architectural Rationale
Runtime & Framework	Node.js & Express.js	Event-driven non-blocking I/O capable of handling concurrent API traffic during registration surges.
Database Layer	MongoDB with Mongoose ODM	Flexible document schemas for varied event dynamic attributes and rapid query indexing.
Auth & Security	JWT, bcryptjs, Helmet, Cors	Stateless token-based authentication, salted password hashing, and HTTP header security.
Data Validation	express-validator	Strict request payload sanitization to prevent NoSQL injection and malformed inputs.

3. SYSTEM FEATURES & KEY DESIGN DECISIONS

- **Role-Based Access Control (RBAC):** Strict segregation between regular `Student` users and authorized `Event Organizer / Admin` accounts.
- **Database Querying & Pagination:** Integrated pagination (limit/page), text search, and category filter parameters for event feeds.
- **Stateless API Architecture:** Clean separation of concerns with Controller-Service-Repository pattern.

4. API ENDPOINT SPECIFICATIONS

Method	Endpoint Route	Description	Auth Level
POST	`/api/v1/auth/register`	Register new student or organizer account	Public
POST	`/api/v1/auth/login`	Authenticate user & issue JWT token	Public
GET	`/api/v1/events`	Fetch all active campus events with filters	Public
POST	`/api/v1/events`	Create new campus event entry	Organizer / Admin
PUT	`/api/v1/events/:id`	Update event details or venue schedules	Event Owner
POST	`/api/v1/registrations`	Register logged-in student for event	Authenticated User

5. MONGODB SCHEMA PREVIEW (EVENT MODEL)

```
const EventSchema = new mongoose.Schema({ title: { type: String, required: true, trim: true }, description: { type: String, required: true }, category: { type: String, enum: ['Coding', 'Design', 'Robotics', 'Gaming'], required: true }, venue: { type: String, required: true }, eventDate: { type: Date, required: true }, capacity: { type: Number, required: true }, organizer: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true } }, { timestamps: true });
```

6. SUBMISSION COMPLIANCE & GITHUB CHECKLIST

- **Public GitHub Repository:** Clean git commit history with descriptive messages, zero committed secret keys (`.env` in `.gitignore`).
- **Comprehensive README.md:** Features installation steps, API documentation links, environment variable guide, and architecture overview.
- **API Documentation:** Exported Postman Collection included in root directory for seamless reviewer evaluation.

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